

Reasoning Behavior Analysis Report

EXPERIMENTAL SETUP

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Data Folder: data/AIME2025__R10/deepseek/p2
Total Items: 300
Total Strategies: 9

OVERALL ACCURACY

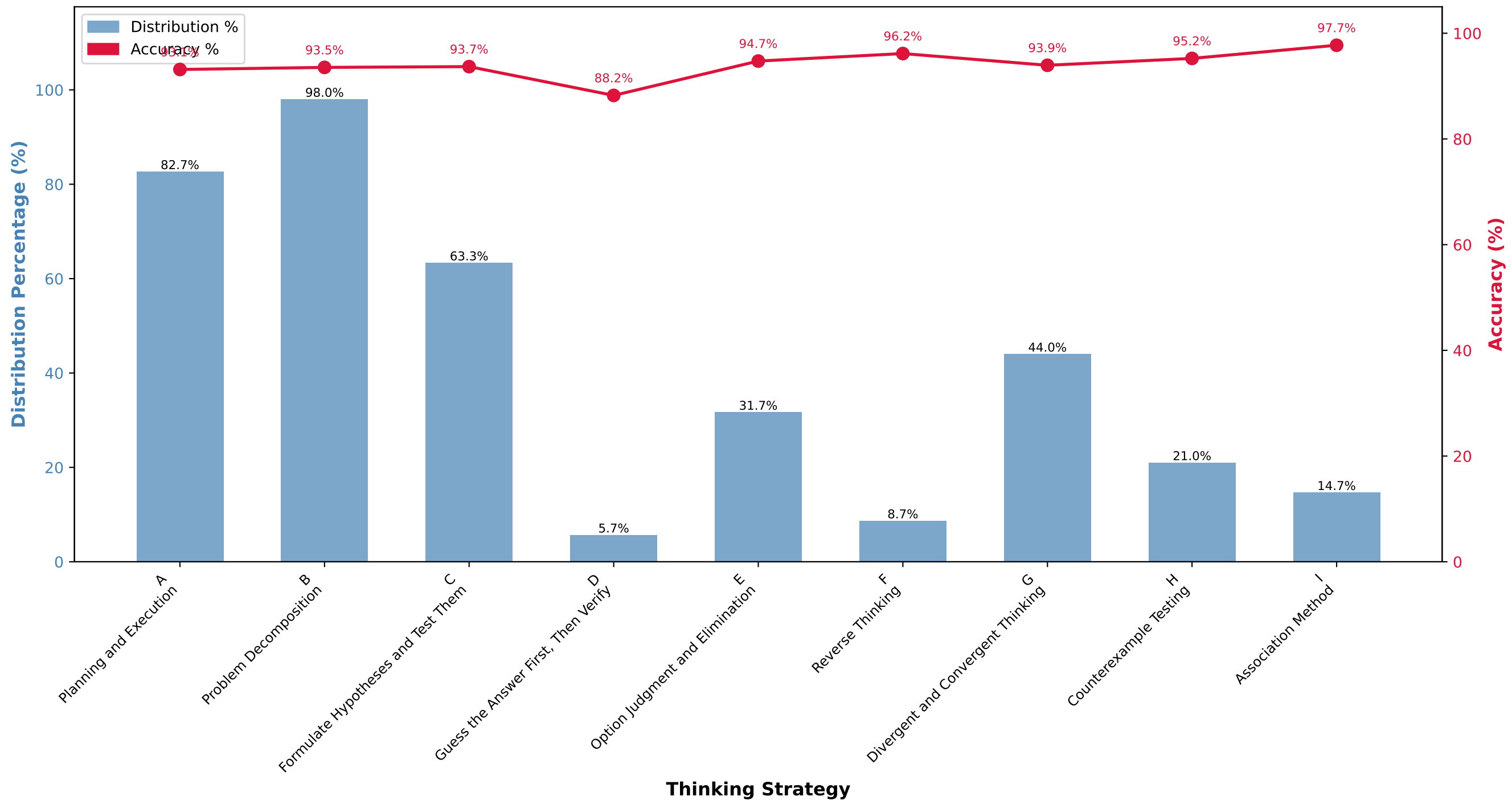
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Total Problems: 300
Correct: 281
Incorrect: 19
Overall Accuracy: 93.67%

THINKING STRATEGY DISTRIBUTION

- =====
- A: Planning and Execution
Distribution: 82.7%
Accuracy: 93.15% (231/248)
 - B: Problem Decomposition
Distribution: 98.0%
Accuracy: 93.54% (275/294)
 - C: Formulate Hypotheses and Test Them
Distribution: 63.3%
Accuracy: 93.68% (178/190)
 - D: Guess the Answer First, Then Verify
Distribution: 5.7%
Accuracy: 88.24% (15/17)
 - E: Option Judgment and Elimination
Distribution: 31.7%
Accuracy: 94.74% (90/95)
 - F: Reverse Thinking
Distribution: 8.7%
Accuracy: 96.15% (25/26)
 - G: Divergent and Convergent Thinking
Distribution: 44.0%
Accuracy: 93.94% (124/132)
 - H: Counterexample Testing
Distribution: 21.0%
Accuracy: 95.24% (60/63)
 - I: Association Method
Distribution: 14.7%
Accuracy: 97.73% (43/44)

Thinking Strategy Distribution and Accuracy



Example for Strategy A: Planning and Execution

PROBLEM

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N/A

GROUND TRUTH ANSWER

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588

MODEL RESPONSE

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The area of heptagon $AFNBCEM$ is found to be 588, which is equal to the area of triangle ABC .

The area of triangle ABC is calculated as follows:

- $AB = AD + DE + EB = 4 + 16 + 8 = 28$
- The height from C to AB is determined from the area of quadrilateral $DEGF$, which is given as 288.
- Using coordinates with A at $(0,0)$, B at $(28,0)$, and C at (p,q) , the area of $DEGF$ is derived as $\frac{48|q|}{7} = 288$, so $|q| = 42$.
- Thus, the area of triangle ABC is $\frac{1}{2} \times AB \times |q| = \frac{1}{2} \times 28 \times 42 = 588$.

The shoelace formula applied to the vertices of heptagon $AFNBCEM$ in order (A, F, N, B, C, E, M) yields an area of 588, confirming that the h...

CHAIN OF THOUGHT (First 1500 chars)

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I need to find the area of this heptagon $AFNBCEM$. First, I should understand the setup. We have triangle ABC with points on AB and AC . On AB , from A to B , we have points A, D, E, B with $AD=4, DE=16, EB=8$. So $AB = AD + DE + EB = 4 + 16 + 8 = 28$.

On AC , from A to C , we have points A, F, G, C with $AF=13, FG=52, GC=26$. So $AC = AF + FG + GC = 13 + 52 + 26 = 91$.

Now, M is the reflection of D through F . Since F is on AC , and D is on AB , reflecting D over F should give me a point M . Similarly, N is the reflection of G through E , so reflecting G over E should give N .

We're given that quadrilateral $DEGF$ has area 288. $DEGF$ is a quadrilateral with points D, E, G, F . From the order, on AB , we have A, D, E, B , and on AC , we have A, F, G, C . So D and E are on AB , F and G are on AC . So $DEGF$ is a quadrilateral formed by points D, E, G, F . Probably, it's connecting D to E to G to F and back to D . Since E and G are not directly connected, but we connect them, so it might be a cross quadrilateral or something. But since D and E are on AB , and F and G are on AC , and we connect D to F , E to G , etc., but the quadrilateral is $D-E-G-F$, so from D to E on AB , then E to G , then G to F on AC ? No, G to F is on AC , but since F is between A and G , G to F would be backwards on AC , but that's fine. Then F to D , which is between AB and AC .

Actually, let's think about the shape. Points D and E are on AB , points F and G are on AC . So when we form quadrilateral $DEGF$, we connect D to E , which is along AB , then...

Example for Strategy B: Problem Decomposition

PROBLEM

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N/A

GROUND TRUTH ANSWER

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821

MODEL RESPONSE

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The probability that the last word listed contains the letter G is $\frac{128}{693}$, where $m = 128$ and $n = 693$ are relatively prime positive integers. Thus, $m + n = 128 + 693 = 821$.

The solution involves considering two cases where G is in the pair with the largest smaller letter (max-S pair). Case 1 is when G is the smaller letter in the max-S pair, and case 2 is when G is the larger letter in the max-S pair. The number of favorable pairings is 1920, and the total number of pairings is 10395. The probability simplifies to $\frac{128}{693}$.

$\boxed{821}$

CHAIN OF THOUGHT (First 1500 chars)

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I need to find the probability that the last word in the list contains the letter G. First, I should understand the setup. There are twelve letters from A to L, and they're paired randomly into six pairs. Then, for each pair, we arrange the two letters in alphabetical order to form a two-letter word. So, for example, if we have pairs like (A,B), (C,J), (D,G), etc., we arrange each pair so that the smaller letter comes first.

After that, we list these six words alphabetically. Since the words are two letters each, we list them based on the first letter, I assume, because if the first letters are the same, we look at the second, but since all first letters are different? No, the first letters could be the same if pairs have the same starting letter, but since we have pairs from A to L, and we're pairing them, the first letters of the words might not be unique because if we have a pair like (A,B) and (A,C), but wait, no, because each letter is used only once, so all first letters will be unique? Let's see.

When we form a word from a pair, we always put the smaller letter first, so for any pair, the first letter of the word is always the smaller one. Since all letters are distinct, the first letters of the words will be distinct only if each pair has a different smaller letter, but since we have six pairs, and twelve letters, the first letters are the six smaller letters from each pair, and since all letters are used, the set of first letters is exactly the set of the six small...

Example for Strategy C: Formulate Hypotheses and Test Them

PROBLEM

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N/A

GROUND TRUTH ANSWER

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204

MODEL RESPONSE

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The disk is initially divided into 4 regions by two perpendicular diameters. With the addition of 25 chords, each drawn between two random points on the perimeter in different quadrants, the total number of lines is 27.

The expected number of regions formed by these lines is given by the formula $1 + C + I$, where C is the number of lines and I is the number of interior intersections. Here, $C = 27$, so the formula becomes $1 + 27 + I = 28 + I$.

The expected number of interior intersections $E[I]$ is the sum of the probabilities of intersection for all pairs of lines:

- There is 1 pair of diameters, which always intersect at the center, contributing 1 .
- There are $2 \times 25 = 50$ pairs of diameter and chord. For each such pair, the probability of...

CHAIN OF THOUGHT (First 1500 chars)

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First, Alex starts by dividing the disk into four quadrants with two perpendicular diameters. So, initially, there are 4 regions from these two lines.

Then, he draws 25 more line segments. Each segment is drawn by selecting two random points on the perimeter in different quadrants and connecting them. I need to find the expected number of regions after all these segments are drawn.

The total number of line segments is 27: 2 diameters and 25 random chords. But the diameters are fixed, and the chords are random.

I recall that for lines or chords in a circle, the number of regions can be found using Euler's formula or other combinatorial methods. But since the chords are random, I need to find the expected number.

I remember that for n points on a circle, if you connect all pairs with chords, the number of regions is given by a formula, but here we're not connecting all pairs; we're drawing random chords between points on the perimeter, and importantly, each chord is between two points in different quadrants.

The quadrants are defined by the two diameters, so the perimeter is divided into four arcs. Each point on the perimeter belongs to one quadrant, and we're only drawing chords between points in different quadrants.

But for the purpose of counting regions, the key is how the chords intersect each other and the existing diameters.

I should think about the total number of intersections and how they affect the number of regions.

There's a known formula for the maximum nu...

Example for Strategy D: Guess the Answer First, Then Verify

PROBLEM

N/A

GROUND TRUTH ANSWER

60

MODEL RESPONSE

Step 1. Understand the Problem

We have a convex pentagon $ABCDE$ with side lengths:

- $AB = 14$
- $BC = 7$
- $CD = 24$
- $DE = 13$
- $EA = 26$

and interior angles:

- $\angle B = 60^\circ$
- $\angle E = 60^\circ$

For any point X in the plane, define:

$$f(X) = AX + BX + CX + DX + EX$$

We aim to find the minimum value of $f(X)$, which can be expressed as:

$$f_{\min} = m + n\sqrt{p}$$

where (m, n) are positive integers and (p) is square-free. Then compute:

$$m + n + p$$

Step 2. Key Observations

The given side lengths and angles lead to two right triangles:

- **Triangle ABC :**
 $AB = 14, BC = 7, \angle B = 60^\circ$
Using the Law of Cosines:
 $AC^2 = 14^2 + 7^2 - 2 \cdot 14 \cdot 7 \cdot \cos(60^\circ) = 14^2 + 7^2 - 98 = 14^2 - 49 = 98 - 49 = 49$
 $AC = 7$

CHAIN OF THOUGHT (First 1500 chars)

We are given: "Let $ABCDE$ be a convex pentagon with $AB=14, BC=7, CD=24, DE=13, EA=26$, and $\angle B = \angle E = 60^\circ$. For each point X in the plane, define $f(X) = AX + BX + CX + DX + EX$. The least possible value of $f(X)$ can be expressed as $m + n\sqrt{p}$, where m and n are positive integers and p is not divisible by the square of any prime. Find $m + n + p$."

Interpretation: We have pentagon vertices A, B, C, D, E . Given side lengths: $AB=14, BC=7, CD=24, DE=13, EA=26$. And angles at B and E are 60° .

We want to minimize $f(X) = \text{sum of distances from } X \text{ to vertices } A, B, C, D, E$.

This is like a Fermat point type problem? But careful: The sum is over 5 points. Possibly the minimum is achieved at a point that is a "Torricelli point" if all angles are $> 120^\circ$. But here we have conditions on angles at B and E . Possibly we can transform the pentagon into a path? There is a technique: For minimizing sum of distances to vertices, sometimes if we can "unfold" the polygon along lines, then the minimal sum is the straight line distance between two endpoints of the unfolded path. But careful: $f(X) = AX + BX + CX + DX + EX$. If we can reflect points, then the sum becomes distance from some reflected version of A to some reflected version of E ? But then there are also vertices B, C, D . Possibly we can do: $f(X) = (AX + EX) + (BX + DX) + CX$. Not exactly.

Alternate approach: Possibly we can reduce the problem by noticing that the pentagon is such that we can "straighten"...

Example for Strategy E: Option Judgment and Elimination

PROBLEM

N/A

GROUND TRUTH ANSWER

259

MODEL RESPONSE

Step 1. Understand the Function and the Intersection Condition

The function $f(x)$ is defined piecewise and is periodic with period 4:

$$f(x) = \begin{cases} x & \text{if } x \in [-1, 1) \\ 2 - x & \text{if } x \in [1, 3) \end{cases}$$
$$\text{and } f(x+4) = f(x)$$

The graph consists of line segments forming a "sawtooth" pattern.

We are to find the intersection points of the graph of $f(x)$ with the parabola:

$$x = 34y^2$$

Since $x \geq 0$, we only consider $x \geq 0$. Also, $f(x) \in [-1, 1]$, so we must have $|y| \leq 1$.

Step 2. Set Up the Equations from the Two Branches

Type A: $f(x) = x$

Let $x = 4n + t$, where $t \in [-1, 1)$. Then:

$$f(x) = t \quad \Rightarrow \dots$$

CHAIN OF THOUGHT (First 1500 chars)

We are given: "A piecewise linear periodic function is defined by $f(x) = \begin{cases} x & \text{if } x \in [-1, 1) \\ 2-x & \text{if } x \in [1, 3) \end{cases}$ and $f(x+4) = f(x)$ for all real numbers x . The graph of $f(x)$ has the sawtooth pattern. The parabola $x = 34y^2$ intersects the graph of $f(x)$ at finitely many points. The sum of the y -coordinates of these intersection points can be expressed in the form $\frac{a+b\sqrt{c}}{d}$, where $a, b, c,$ and d are positive integers, $a, b,$ and d have greatest common divisor equal to 1, and c is not divisible by the square of any prime. Find $a+b+c+d$."

Interpretation: f is periodic with period 4. $f(x) = ?$ Actually careful: $f(x)$ defined on $[-1, 1)$ and $[1, 3)$ and then extended periodically. So pattern: $f(x)$ is like a "sawtooth" pattern. Actually: $f(x) = ?$ For $x \bmod 4$, if we take the representative in $[-1, 3)$ then: For $x \bmod 4$ in $[-1, 1)$: $f(x) = x \bmod 4$? Actually careful: The definition: $f(x) = x$ for $x \in [-1, 1)$; $f(x) = 2-x$ for $x \in [1, 3)$. And then $f(x+4) = f(x)$. So it's periodic with period 4. So its graph: It consists of line segments. On each period, say from $x = 4k - 1$ to $4k + 1$? Actually, careful: The pattern: For an integer n , the function on $[4n-1, 4n+1)$ is: $f(x) = ?$ Actually, if we set $x = 4n + t$, where $t \in [-1, 1)$, then $f(x) = f(4n+t) = f(t) = t$. So on these intervals, $f(x) = x - 4n$? Actually: if $t \in [-1, 1)$, then $f(4n+t) = t$. So that line is: $y = t$, and $x = 4n+t$, so $y = x - 4n$. And on the other part: For $x \bmod 4$ in $[1, 3)$, $f(x) = 2 - x \bmod 4$. So if $x = 4n + t$ where $t \in [1, 3)$, then $f(x) = 2 - t$. So $y = 2 - t$, and $x = 4n + t$, so $y = 2 - (x - 4n) = 4n + 2 - x$.

Example for Strategy F: Reverse Thinking

PROBLEM

N/A

GROUND TRUTH ANSWER

77

MODEL RESPONSE

The system consists of two equations:

1. $|25 + 20i - z| = 5$, which represents a circle centered at $(25 + 20i)$ with radius 5.
2. $|z - 4 - k| = |z - 3i - k|$, which represents the perpendicular bisector of the points $(4 + k)$ and $(3i + k)$.

For the system to have exactly one complex solution z , the line must be tangent to the circle. The distance from the center of the circle $(25, 20)$ to the line $(8x - 6y - 8k - 7 = 0)$ must equal the radius 5.

The distance is given by:

$$d = \frac{|8 \cdot 25 - 6 \cdot 20 - 8k - 7|}{\sqrt{8^2 + (-6)^2}} = \frac{|200 - 120 - 8k - 7|}{10} = \frac{|73 - 8k|}{10}$$

Setting $d = 5$:

$$\frac{|73 - 8k|}{10} = 5 \implies |73 - 8k| = 50$$

This gives two cases:

$$- \quad 73 - 8k = 50 \implies 8k = 23 \dots$$

CHAIN OF THOUGHT (First 1500 chars)

I need to solve this problem. It involves complex numbers and conditions for exactly one solution. Let me understand the equations.

First, $|25 + 20i - z| = 5$. This represents a circle in the complex plane centered at $25 + 20i$ with radius 5. So, z lies on this circle.

The second equation is $|z - 4 - k| = |z - 3i - k|$. I notice that k is a real number, and it appears in both terms. Let me simplify this.

$$|z - 4 - k| = |z - 3i - k|$$

I can factor out the k ? Well, since k is real, but z is complex, I should think of this geometrically.

This equation means that the distance from z to $4 + k$ is equal to the distance from z to $3i + k$. But $4 + k$ is a point on the real axis, since k is real, so it's $(4 + k, 0)$. Similarly, $3i + k$ is $(k, 3)$ because $3i$ is on the imaginary axis, so with real part k and imaginary part 3.

Let me denote complex numbers as points in the plane. So let $z = x + iy$, where x and y are real.

$$\text{Then } |z - 4 - k| = |z - 3i - k|.$$

$$z - 4 - k = (x - 4 - k) + i y$$

$$z - 3i - k = (x - k) + i (y - 3)$$

So the equation is:

$$|(x - 4 - k) + i y| = |(x - k) + i (y - 3)|$$

Which means:

$$\sqrt{(x - 4 - k)^2 + y^2} = \sqrt{(x - k)^2 + (y - 3)^2}$$

Squaring both sides:

$$(x - 4 - k)^2 + y^2 = (x - k)^2 + (y - 3)^2$$

Expand both sides:

$$\text{Left: } (x - k - 4)^2 + y^2 = (x - k)^2 - 8(x - k) + 16 + y^2 \text{? Let me compute properly.}$$

Set $u = x - k$, to simplify. Since k is real, u is real.

$$\text{So } (u - 4)^2 + y^2 = u^2 + (y - 3)^2$$

Expand:

$$\text{Left: } u^2 - 8u + 16 + y^2$$

$$\text{Right: } u^2 + y^2 - 6y + 9$$

So:

$$u^2 - \dots$$

Example for Strategy G: Divergent and Convergent Thinking

PROBLEM

N/A

GROUND TRUTH ANSWER

588

MODEL RESPONSE

The area of heptagon AFNBCEM is found to be 588, based on coordinate geometry and the shoelace formula. The calculation shows that the area is independent of the angle between sides AB and AC, as the terms involving cosine cancel out.

The points are defined as follows:

- A is at (0, 0).
- B is at (28, 0) since $AB = 28$.
- C is at $(91 \cos \theta, 91 \sin \theta)$, with $\sin \theta = 6/13$ from the given area of quadrilateral DEGF.
- F is on AC at $(13 \cos \theta, 6)$ since $AF = 13$ and $\sin \theta = 6/13$.
- N is the reflection of G through E, where G is at $(65 \cos \theta, 30)$ and E is at (20, 0), so N is at $(40 - 65 \cos \theta, -30)$.
- E is at (20, 0).
- M is the reflection of D through F, where D is at (4, 0) and F is at $(13 \cos \theta, 6)$, so M is at $(26 \cos \theta - 4, 12)$.

The heptagon vertices are listed in order: A, F, N, B, C, E, M. U...

CHAIN OF THOUGHT (First 1500 chars)

I need to find the area of this heptagon AFNBCEM. First, I should understand the setup. We have triangle ABC with points on AB and AC. On AB, from A to B, we have points A, D, E, B in that order with $AD=4$, $DE=16$, $EB=8$. So $AB = AD + DE + EB = 4 + 16 + 8 = 28$.

On AC, from A to C, we have points A, F, G, C with $AF=13$, $FG=52$, $GC=26$. So $AC = AF + FG + GC = 13 + 52 + 26 = 91$.

Now, M is the reflection of D through F. Since F is on AC, and D is on AB, reflecting D over F should give me a point M. Similarly, N is the reflection of G through E. E is on AB, G is on AC, so reflecting G over E should give N.

We're given that quadrilateral DEGF has area 288. DEGF is a quadrilateral with points D, E, G, F. Since D and E are on AB, and F and G are on AC, this quadrilateral is probably inside the triangle.

I need to find the area of heptagon AFNBCEM. That seems like a seven-sided figure with points A, F, N, B, C, E, M. But let's list the points: A, F, N, B, C, E, M. That's seven points, so heptagon.

To find the area, I probably need to find the area of triangle ABC and subtract the areas of the parts not included in the heptagon, or something like that.

Since we have reflections, I might use properties of reflections or coordinates.

I should set up a coordinate system to make this easier. Let me place point A at the origin for simplicity.

Set A at (0,0). Since AB and AC are two sides, I can put B on the x-axis and C on the y-axis, but that would make triangle ABC right-angled, which m...

Example for Strategy H: Counterexample Testing

PROBLEM

N/A

GROUND TRUTH ANSWER

16

MODEL RESPONSE

The valid distributions of flavors that satisfy the conditions are (5,3,1), (6,2,1), and (4,3,2), where the numbers represent the counts of chocolate, vanilla, and strawberry, respectively.

For each distribution, the number of ways to assign flavors to the distinct players is calculated using multinomial coefficients:

- For (5,3,1): $\frac{9!}{5!3!1!} = 504$
- For (6,2,1): $\frac{9!}{6!2!1!} = 252$
- For (4,3,2): $\frac{9!}{4!3!2!} = 1260$

The total number of assignments (N) is the sum:

$$504 + 252 + 1260 = 2016$$

The remainder when (N) is divided by 1000 is:

$$2016 \bmod 1000 = 16$$

Thus, the remainder is 16.

$\boxed{16}$

CHAIN OF THOUGHT (First 1500 chars)

I need to find the number of ways to assign flavors to the nine players such that each flavor is used at least once, and the number of chocolate is greater than vanilla, which is greater than strawberry. Then I need to find that number modulo 1000.

First, since there are three flavors and nine players, and each flavor has at least one, I need to consider the possible distributions of the flavors. Let C be the number of chocolate, V be vanilla, and S be strawberry. We have $C > V > S$, and $C + V + S = 9$, with each at least 1.

Since $C > V > S$, and all are integers, S must be at least 1, V at least 2, and C at least 3, but since $C > V$, and $V > S$, it might be tighter. Let me list all possible triplets (C,V,S) such that $C + V + S = 9$, $C > V > S$, and $S \geq 1$, $V \geq 2$, $C \geq 3$.

I'll start with S=1. Then $V > S$, so $V \geq 2$, and $C > V$, so $C \geq 3$. $C + V + 1 = 9$, so $C + V = 8$. Since $C > V$, possible pairs (C,V) with $C + V = 8$ and $C > V$: C=5,V=3; C=6,V=2 but V=2, S=1, so $V > S$, ok? C=6,V=2, but V=2 and S=1, so $V > S$, yes. C=7,V=1 but V=1 not greater than S=1, since V must be greater than S, so V cannot be 1 if S=1. Similarly, C=4,V=4 but C not greater than V. So for S=1, $C + V = 8$, $C > V$, so possible: C=5,V=3; C=6,V=2; C=7,V=1 but V=1 not greater than S=1, so invalid. So only C=5,V=3 or C=6,V=2.

C=6,V=2,S=1: here $V=2 > S=1$, yes. And $C=6 > V=2$, yes.

C=7,V=1,S=1: but $V=1$ not greater than S=1, so no.

So for S=1, we have (5,3,1) and (6,2,1).

Now S=2. Then $V > S$, so $V \geq 3$, and $C > V$, so $C \geq 4$. $C + V + 2 = 9$, so $C + V = 7$. $C > V$, so possible:...

Example for Strategy I: Association Method

PROBLEM

N/A

GROUND TRUTH ANSWER

735

MODEL RESPONSE

Step 1. Reformulate the Problem

We want to count the number of ordered triples $((a, b, c))$ with

$1 \leq a, b, c \leq 3^6 = 729$

such that

$a^3 + b^3 + c^3 \equiv 0 \pmod{3^7}.$

A key observation: if two numbers are congruent modulo (3^6) , then their cubes are congruent modulo (3^7) . Thus, we can view $((a, b, c))$ as elements of the ring $(\mathbb{Z}/3^6)$.

The condition becomes:

$x^3 + y^3 + z^3 \equiv 0 \pmod{3^7}, \quad \text{where } x, y, z \in \mathbb{Z}/3^6.$

Step 2. A Useful Idea: Density of Solutions

Let's consider a smaller case to detect a pattern.

Suppose we want to count the number of triples $((a, b, c))$ with

$1 \leq a, b, c \leq 3^2 = 9$

such that

$a^3 + b^3 + c^3 \equiv 0 \pmod{3^3} = 27.$

CHAIN OF THOUGHT (First 1500 chars)

We are given: "Let N denote the number of ordered triples of positive integers (a, b, c) such that $a, b, c \leq 3^6$ and $a^3 + b^3 + c^3$ is a multiple of 3^7 . Find the remainder when N is divided by 1000 ."

We have: $a, b, c \in \{1, 2, \dots, 3^6\}$. Condition: 3^7 divides $a^3 + b^3 + c^3$.

Let's denote: $n = 6$, so numbers are from 1 to 3^6 . And condition: 3^7 divides sum of cubes.

We want to count number of ordered triples (a, b, c) such that $a^3 + b^3 + c^3 \equiv 0 \pmod{3^7}$.

Observation: Write numbers mod powers of 3. But careful: The numbers are from 1 to 3^6 . They are not necessarily all residues mod something? But we can count by considering representation mod 3^7 ? Actually condition is mod 3^7 . But a, b, c are numbers up to 3^6 . So they are not complete residue system mod 3^7 ? Actually, numbers from 1 to 3^6 : They are all numbers that are not multiples of 3? Actually, wait: $3^6 = 729$. So numbers are $1, \dots, 729$. They are not all mod 3^7 because mod 3^7 , numbers go from 0 to 2186. But here we only have numbers that are ≤ 729 . So they are a subset of residues mod 3^7 . But maybe we can use a method: Count for each a mod something? Alternatively, we can break into cases based on the 3-adic valuation of a, b, c .

Let $v_3(x)$ denote exponent of 3 dividing x . Then a^3 : $v_3(a^3) = 3 v_3(a)$. So condition: $a^3 + b^3 + c^3$ is divisible by 3^7 . That means $v_3(a^3 + b^3 + c^3) \geq 7$.

We consider the numbers a, b, c . They are positive integers $\leq 3^6$. Write $a = 3^k \cdot a_0$, where a_0 is not divisible by 3. ...